

Table 3 Best Practice Protocol for Undertaking Whole Body Composition Assessment by DXA in Athletes and Active People

General			Implications	
• DXA assessments should be undertaken in conjunction with surface anthropometry to better understand the relationship between the two measurements. • Near nude body mass should also be measured on a calibrated scale for a quick and simple reliability cross-check against DXA-derived estimate of total mass. • Longitudinal monitoring of athletes should be undertaken using the same DXA machine and, ideally, the same technician. • Each facility is encouraged to calculate their machine- and population-specific technical error of measurement (e.g., % coefficient of variation, least significant change). • The frequency of measurement of whole body composition in an individual should be determined according to the likelihood that any change exceeds the measurement error, as well as local limitations of radiation exposure (e.g., requirements of Human Ethics Committees or Radiation Safety authorities)			• High subject and technician burden (early mornings) • Limited number of scans per morning (difficult to scan large groups) • Limits use of DXA machine for monitoring body composition to mornings (high expense for short periods of use)	
Stage	Type of Error	Point of Error	Best Practice Protocol	Notes
Prescanning	Technical	Subject's clothing	• Minimal clothing (e.g., underwear and wire-free crop top with jewelry removed).	Minimal clothing provides better visualization of subjects' positioning on the scanning bed (spine alignment and limb placement). Clothing should also be free of chlorine and salt since these factors may alter attenuation ratios and subsequent estimates of body composition.
	Biological	Subject presentation—acute and chronic hydration status	• Fasted with no fluid intake on morning of scan • No exercise on morning of scan • Optimized hydration status (collect upon waking urine specific gravity on assessment mornings to check) • Bladder voided	USG maybe useful for longitudinal monitoring of body composition. Although unclear, major alterations in hydration may be at least help explain incongruent body composition estimates. Other unknown sources of error may include menstrual cycle, glycogen stores, gastrointestinal gas/content or use of supplements that affect intramuscular solutes (e.g., creatine, β -alanine).

(continued)

Table 3 (continued)

Stage	Type of Error	Point of Error	Best Practice Protocol	Notes
Scanning	Technical	Machine's inherent "technical noises"	• Undertake routine quality control and quality assurance system in place to detect any "drift" in the system.	Each facility is encouraged to calculate their own machine's technical reliability from repositioning of subjects (e.g., % coefficient of variation), to assist in inferring if a real change has occurred.
		Technician's positioning of subject on the scanner	• Standardized positioning technique with the possible use of positioning aids.	Food and hand positioning aids are designed to standardize the positioning of the subject on the scanning bed and to maximize the consistency of scans within- and between-subjects. Our experience suggests that they aid subject comfort in maintaining the desired scanning position and improve the apparent reproducibility of results.
Analysis of regional composition	Technical	Manual demarcation of segmental lines on the body	• Check regional analysis by the software. • Have standardized regional composition analysis protocol.	The experience of the technician may affect the consistency of scans. If possible, calculate the intra- and intertechnician variability specific to the laboratory.
		Consistency of scans within- and between-subjects.	• Ideally, have one technician undertake and analyze the scan over time.	

Table 4 Information that Should be Included in Reports of Study Methodology When Using DXA to Estimate Body Composition in Athletes and Active People

Themes	Reporting Conditions
Subject presentation	• Fasting state • Resting state • Hydration status • Bladder voided or not • Clothing
DXA instrument	• Brand • Model • Beam configuration • Analysis software version • Scanning mode • Machine's technical reliability calculated from repositioning of subjects (e.g., % coefficient of variation)
Positioning and analysis protocol	• Description of how the subjects are positioned on the scanning bed including hand and feet positioning (and with/without the use of positioning aids) • If regional composition is reported—report the analysis technique (e.g., fully automatic by the software, automatic by the software and confirmed by the technician, or manual analysis by the technician) • Detailed description of any custom region of interest • Specific techniques used to scan subpopulations (e.g., tall and/or broad subjects, athletes with disability)
Number of technicians undertaking scans (especially for longitudinal monitoring)	• The number of technicians involved • Each technician's role (e.g., scanning, analyzing or both scanning and analysis)